

The Assignment Problem

The Assignment Problem

- In many business situations, management needs to assign - personnel to jobs, - jobs to machines, - machines to job locations, or - salespersons to territories.
- Consider the situation of assigning n jobs to n machines.
- When a job i ($=1,2,\dots,n$) is assigned to machine j ($=1,2,\dots,n$) that incurs a cost C_{ij} .
- The objective is to assign the jobs to machines at the least possible total cost.

The Assignment Problem

- This situation is a special case of the Transportation Model And it is known as the *assignment problem*.
- Here, jobs represent “sources” and machines represent “destinations.”
- The supply available at each source is 1 unit And demand at each destination is 1 unit.

The Assignment Problem

	Machine					Source
		1	2		n	
Job	1	C11	C12		C1n	1
	2	C21	C22		C2n	1
	.	.	.		.	.
	.	.	.		.	.
	.	.	.		.	.
	n	Cn1	Cn2		Cnn	1
Destination		1	1		1	

The assignment model can be expressed mathematically as follows:

$X_{ij} = \begin{cases} 0, & \text{if the job } j \text{ is not assigned to machine } i \\ 1, & \text{if the job } j \text{ is assigned to machine } i \end{cases}$

The Assignment Problem Example

- Ballston Electronics manufactures small electrical devices.
- Products are manufactured on five different assembly lines (1,2,3,4,5).
- When manufacturing is finished, products are transported from the assembly lines to one of the five different inspection areas (A,B,C,D,E).
- Transporting products from five assembly lines to five inspection areas requires different times (in minutes)

The Assignment Problem Example

Assembly Line	Inspection Area				
	A	B	C	D	E
1	10	4	6	10	12
2	11	7	7	9	14
3	13	8	12	14	15
4	14	16	13	17	17
5	19	11	17	20	19

Under current arrangement, assignment of inspection areas to the assembly lines are 1 to A, 2 to B, 3 to C, 4 to D, and 5 to E.

This arrangement requires $10+7+12+17+19 = 65$ man minutes.

Hungarian Method Example

		Machine			
		1	2	3	
Job	1	5	7	9	1
	2	14	10	12	1
	3	15	13	16	1
		1	1	1	

Step 1: Select the smallest value in each row.
Subtract this value from each value in that row

Step 2: Do the same for the columns that do not have any zero value.

Hungarian Method Example

		Machine		
		1	2	3
Job	1	5	7	9
	2	14	10	12
	3	15	13	16

		Machine		
		1	2	3
Job	1	0	2	4
	2	4	0	2
	3	2	0	3

	Machine		
	1	2	3
1	0	2	2
2	4	0	0
3	2	0	1

If not finished, continue with other columns.

Hungarian Method Example

Step 3: Assignments are made at zero values.

- Therefore, we assign job 1 to machine 1; job 2 to machine 3, and job 3 to machine 2.
- Total cost is $5+12+13 = 30$.
- It is not always possible to obtain a feasible assignment as in here.

Hungarian Method Example 2

	1	2	3	4
1	<u>1</u>	4	6	3
2	9	<u>7</u>	10	9
3	<u>4</u>	5	11	7
4	8	7	8	<u>5</u>

	1	2	<u>3</u>	4
1	0	3	<u>5</u>	2
2	2	0	<u>3</u>	2
3	0	1	<u>7</u>	3
4	3	2	<u>3</u>	0

	1	2	3	4
1	<u>0</u>	3	2	2
2	2	<u>0</u>	0	2
3	0	1	4	3
4	3	2	<u>0</u>	0

Hungarian Method Example 2

- A feasible assignment is not possible at this moment.
- In such a case, The procedure is to draw a *minimum* number of *lines* through some of the rows and columns, *Such that all zero values are crossed out.*

Hungarian Method Example 2

	1	2	3	4
1	0	3	2	2
2	2	0	0	2
3	0	1	4	3
4	3	2	0	0

The next step is to select the smallest uncrossed out element. This element is *subtracted from every uncrossed out element* and *added to every element at the intersection* of two lines.

	1	2	3	4
1	<u>0</u>	2	1	1
2	3	0	<u>0</u>	2
3	0	<u>0</u>	3	2
4	4	2	0	<u>0</u>

Hungarian Method Example 2

- We can now easily assign to the zero values. Solution is to assign (1 to 1), (2 to 3), (3 to 2) and (4 to 4).
- If drawing lines do not provide an easy solution, then we should perform the task of drawing lines one more time.
- Actually, we should continue drawing lines until a feasible assignment is possible.